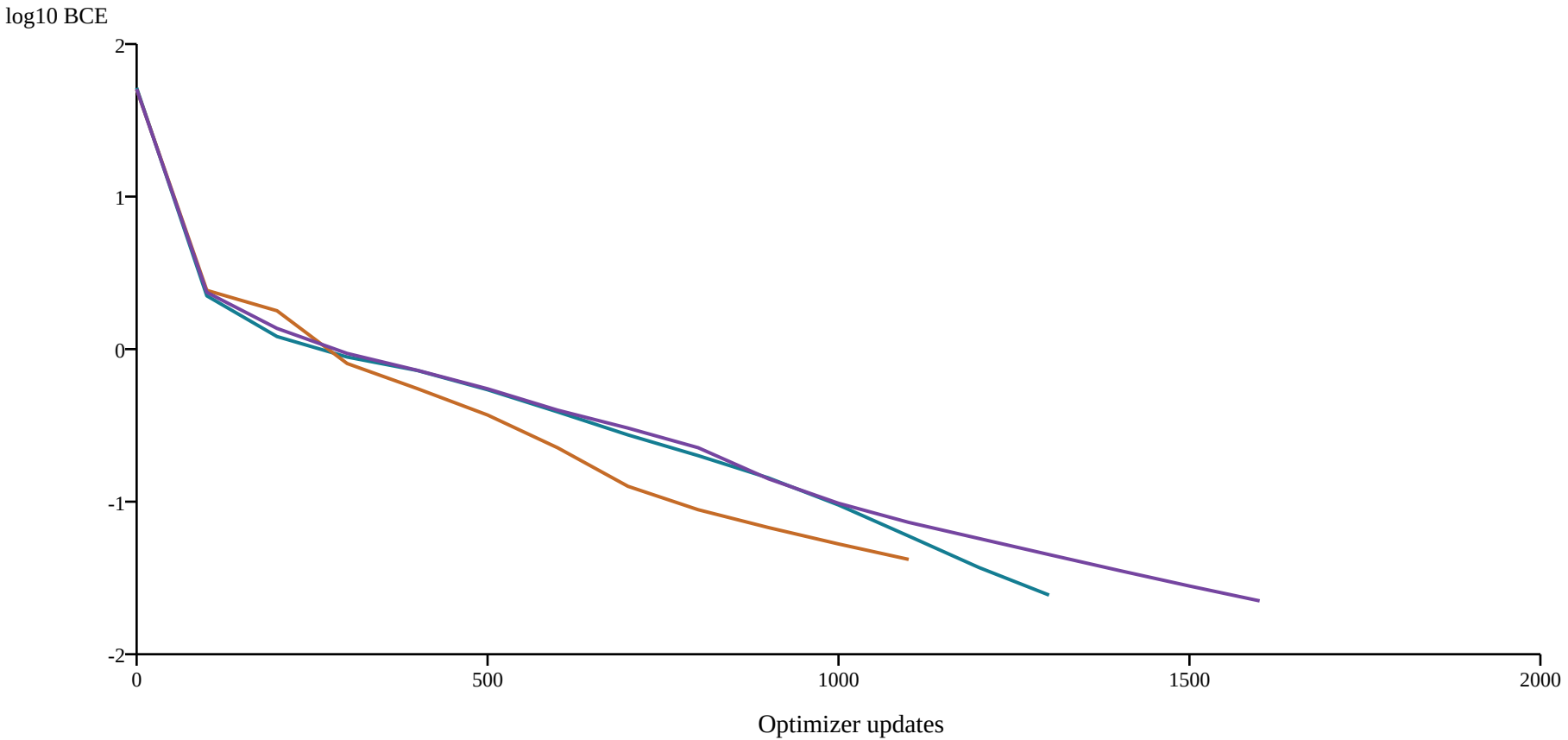Fixed small set: weighted BCE during optimization